

(1) Example of a Disruptive Technology

An example of a disruptive technology described by Christensen is digital streaming video popularized by Netflix.

Netflix's streaming service initially offered lower performance than traditional DVD rentals, which was its bread-and-butter at the time and also had strong competition by rental stores like Blockbuster. The catalog was limited, video quality was often poor, and new releases lagged behind other methods. As a result, the service initially appealed to non-core customers. These consumers made these tradeoffs for convenience and value.

This fits Christensen's disruption model because incumbents like Blockbuster focused on sustaining their core business. They prioritized (and likely gained) customers who wanted new releases immediately and preferred high-quality media.

During this time of stagnation by brick-and-mortar firms, streaming technology gradually improved due to better internet infrastructure, device compatibility, and growing licensing agreements. Over time, streaming ultimately became good enough for mainstream consumers, leading millions of customers to abandon physical rentals entirely.

This trajectory of starting with a lower-tier product targeting willing customers and gradually improving it, eventually was enough to displace incumbents. This closely matches Christensen's theory of disruptive innovation.

(2) Example of a Sustaining Technology

An example of a sustaining technology within Clayton M. Christensen's definition is the evolution of Wi-Fi. This technology has had incremental improvements overtime and standards changed from Wi-Fi 5 to Wi-Fi 6 among others.

Sustaining innovations improve existing products within features that mainstream customers already value. For Wi-Fi this is speed, reliability, and capacity. New generations increased throughput, reduced latency, and increased security greatly over a product that was functionally the same but lacking these features in previous iterations.

Also, these improvements do not necessarily create a new market or target different users. Instead, they improve upon a widely adopted technology used by businesses, homes, and infrastructure. Organizations upgrade their networking equipment as needed to support increasing requirements which don't necessarily disrupt the existing market.

Incumbent firms like Cisco and PaloAlto dominate this space by continuously refining their products. They are able to invest heavily in R&D to ensuring swift compatibility

with new standards and delivering incremental but meaningful improvements to customers.

For these reasons, Wi-Fi represents a sustaining technology by following existing performance and serves current mainstream users rather than displacing them with low-cost alternatives.

(3) Technology That Did Not Start at the Low End but Became a Major Player

An example of a technology that did not originate at the low-performance end but became a major competitor is electric vehicles (EVs) and manufacturers starting with Tesla.

Traditional auto manufacturers such as Toyota and Ford have long dominated the market with internal combustion and occasionally hybrid systems. However, Tesla entered the market without following Christensen's disruptive path of targeting low-end consumers. Tesla launched as a high-performance sports car aimed at wealthy early adopters. The Roadster was expensive and advanced instead of a simple or low-cost EV.

Starting at the premium end of the market is opposite Christensen's model of beginning with inferior performance for low-end "change-willing" customers. Tesla's first vehicles instead emphasized innovative battery technology and premium feeling software.

Tesla's growth involved gathering the revenue of selling niche luxury vehicles to develop their products to a point where they could eventually become mainstream products. This consistent and aggressive R&D by Tesla engineers allowed them to bring cost down and convince the public that electric vehicles were more than just a fad. This eventually led to Tesla targeting traditional market consumers. It also caught the attention of traditional auto manufacturers, which only accelerated this process.

Several other factors contributed to Tesla's rapid growth. Advances in lithium-ion battery technology significantly improved and along with it came range and performance, critical aspects of a vehicle. Second, Tesla's direct-to-consumer sales model and software-focused vehicle architecture differentiated from traditional automakers. Finally, environmental concerns and government incentives accelerated federal approval and consumer adoption.

Tesla hasn't so much displaced traditional automakers, but instead has become one of the major players, especially among EVs. Other manufacturers have responded by investing heavily in EV platforms; however, they've not had as much success.

Tesla has demonstrated that successful disruption does not always follow Christensen's low-end path.

(4) Technology That Displaced Incumbents Without Starting at the Low End

A technology that did not originate at the low end but eventually displaced incumbents is digital photography. This replaced traditional film dominated by Kodak and Fujifilm.

Digital photography was typically reserved for research laboratories and early professionals. Low-end consumer markets were long priced out and disinterested by the technology which existed as early as 1975.

Initially, digital photography was not targeted at the low end of the market. Early digital cameras were professional tools used by news agencies, researchers, and specialized industries where instant images were valuable. These cameras were high-cost, often thousands of dollars, and were not accessible to market consumers.

Over time, advances in sensors, image processing, and storage all dramatically improved. These technological advances greatly increased digital camera performance. Following this, prices fell and image quality improved. Digital cameras became attractive to mainstream consumers who valued the features provided to them by the tech.

The growth trajectory accelerated in the late 90s and early 2000s. Consumer adoption increased even faster as digital cameras became integrated into more and more devices and the support for the technology grew with smartphones and social media. The instantaneous nature of digital photography fundamentally changed the need for film.

This shift ultimately displaced the traditional film industry. Demand for photographic film collapsed as consumers transitioned to digital devices. Kodak, which had dominated the global film market for decades, struggled to adapt its business model.

Digital photography is a prime example of a technological transition that did not begin as low-performance but still completely transformed and replaced an existing industry.

(B) Opinion: Applicability of Disruption Theory to the Digital World

Christensen's theory of disruptive innovation can be an important framework for understanding technological change. However, its applicability in the modern economy may be limited in scope and put blinders on someone using it to study digital innovations. Some innovations clearly follow the disruptive pattern of starting with lower performance and moving upmarket. However, many of the most impactful technologies do not follow this trend.

Examples such as Tesla electric vehicles and digital photography display that technological transformations can originate in specialized markets and not exist for a long time at low-end markets. In these cases, they appeal to premium or professional users before expanding to eventually capture mainstream consumers. This growth does not align with Christensen's classic model but is still extremely common.

Additionally, digital technologies can scale rapidly due to network effects. Also, some technologies may thrive or fail strictly because of their distribution channels. These possibilities make competition nonlinear and more complex.

Focusing exclusively on “disruptive technologies” would overlook innovations that have had profound impact in the economy and on society. Things such as sustaining innovations, platform technologies, and high-end products can be as transformative as low-end disruptions.

For this reason, a course on managing digital transformation (such as MDT) should expand beyond Christensen’s definition of disruption and study a broader range of innovation.